

# SQL Interview Questions

Q1. What is SQL?

Answer:

SQL (Structured Query Language) is used to communicate with databases to store, retrieve, update, and delete data.

Q2. Types of SQL Commands?

Answer:

- DDL (Data Definition Language) → CREATE, ALTER, DROP, TRUNCATE
- DML (Data Manipulation Language) → INSERT, UPDATE, DELETE
- DQL (Data Query Language) → SELECT
- TCL (Transaction Control Language) → COMMIT, ROLLBACK, SAVEPOINT
- DCL (Data Control Language) → GRANT, REVOKE

Q3. Difference between DBMS and RDBMS?

Answer:

| Feature   | DBMS           | RDBMS               |
|-----------|----------------|---------------------|
| Structure | Non-relational | Relational (tables) |
| Relations | Not Supported  | Supported           |
| Example   | File System    | MySQL, Oracle       |

Q4. Most Used SQL Commands

Answer:

- SELECT → Retrieve data
- INSERT → Add records
- UPDATE → Modify records
- DELETE → Remove records

Example:

```
SELECT * FROM Employees;
```

**Pro Tip:** SELECT, JOIN, GROUP BY, ORDER BY, and WHERE are asked in almost every SQL interview.

Q5. What is a Table?

Answer:



### Q5. What is a Table?

Answer:

A table is a collection of data organized in rows and columns.  
Each row represents a record and each column represents an attribute.

### Q6. What is a Row and Column?

Answer:

- Row: A single record in a table.
- Column: An attribute or field that holds a specific type of data.

### Q7. What are the data types in SQL?

Answer:

- INT → Integer numbers
- VARCHAR(n) → Variable length string
- CHAR(n) → Fixed length string
- DATE → Date value
- FLOAT/DECIMAL → Decimal numbers
- BOOLEAN → True / False

### Q8. Difference between CHAR and VARCHAR?

Answer:

- CHAR(n): Fixed length. Uses exact n characters.
- VARCHAR(n): Variable length. Uses only required space.

Example:

CHAR(10) → Always 10 characters

VARCHAR(10) → 1 to 10 characters

### Q9. What is NULL?

Answer:

NULL represents missing, unknown or inapplicable data in a table.

**Pro Tip:** NULL is not equal to zero or empty string.  
Use IS NULL to check for NULL values.

### Q10. What is the difference between DELETE, TRUNCATE and DROP?

Answer:

- DELETE → Removes specific rows based on condition.  
It is DML command and can be rolled back.
- TRUNCATE → Removes all rows from table but keeps structure.  
It is DDL command and faster than DELETE.
- DROP → Removes the entire table structure and data.  
It is DDL command and cannot be rolled back.



# SQL Interview Questions

Q11. What is PRIMARY KEY?

Answer:

A primary key is a column (or combination of columns) that uniquely identifies each record in a table.

Q12. What is FOREIGN KEY?

Answer:

A foreign key is a column (or combination of columns) that refers to the primary key in another table. It maintains referential integrity.

Q13. Types of Constraints in SQL?

Answer:

- NOT NULL → Column cannot have NULL values
- UNIQUE → All values in the column must be unique
- PRIMARY KEY → Uniquely identifies each record
- FOREIGN KEY → Ensures referential integrity
- CHECK → Ensures that values satisfy a specific condition
- DEFAULT → Sets a default value for a column

Q14. What is normalization?

Answer:

Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity.

Q15. Types of Normalization?

Answer:

- 1NF → Atomic values (no repeating groups)
- 2NF → Remove partial dependency
- 3NF → Remove transitive dependency
- BCNF → Stronger version of 3NF

Q16. Difference between WHERE and HAVING?

| Feature   | WHERE                                      | HAVING                                                                  |
|-----------|--------------------------------------------|-------------------------------------------------------------------------|
| Used with | All queries                                | Aggregate functions (GROUP BY)                                          |
| Filters   | Rows before grouping                       | Groups after grouping                                                   |
| Example   | SELECT * FROM Emp<br>WHERE Salary > 50000; | SELECT Dept, COUNT(*)<br>FROM Emp GROUP BY Dept<br>HAVING COUNT(*) > 5; |

**Pro Tip:** WHERE filters rows, HAVING filters groups.



# SQL Interview Questions

Q17. What is GROUP BY?

Answer:

GROUP BY clause is used to group rows that have the same values into summary rows, often used with aggregate functions (COUNT, SUM, AVG, etc.).

Q18. What is ORDER BY?

Answer:

ORDER BY is used to sort the result set in ascending (ASC) or descending (DESC) order.

Q19. Difference between UNION and UNION ALL?

Answer:

- UNION → Combines results and removes duplicate rows.
- UNION ALL → Combines results and keeps duplicate rows.

Q20. What is JOIN?

Answer:

JOIN is used to combine rows from two or more tables based on a related column between them.

Types of JOIN:

- INNER JOIN → Returns matching rows from both tables
- LEFT JOIN (or LEFT OUTER JOIN) → Returns all rows from left table and matching rows from right table
- RIGHT JOIN (or RIGHT OUTER JOIN) → Returns all rows from right table and matching rows from left table
- FULL JOIN (or FULL OUTER JOIN) → Returns all rows when there is a match in either left or right table
- CROSS JOIN → Returns Cartesian product of both tables

Q21. What is the difference between INNER JOIN and LEFT JOIN?

Answer:

- INNER JOIN → Returns only matching rows from both tables.
- LEFT JOIN → Returns all rows from left table and matching rows from right table. If no match, NULL is returned for right table columns.

**Pro Tip:** Practice writing queries on real-world scenarios. It helps in understanding joins, filtering and aggregation better.

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# SQL Interview Questions

## Q22. What is a Subquery?

Answer:

A subquery is a query nested inside another query.

It is used to retrieve data that will be used in the main query.

Example:

```
SELECT Name FROM Employees
WHERE DeptID IN (SELECT DeptID FROM Departments);
```

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## Q23. What is the difference between WHERE and HAVING?

Answer:

- WHERE filters rows before grouping.
- HAVING filters groups after grouping.

Example:

```
WHERE: SELECT * FROM Emp WHERE Salary > 50000;
HAVING: SELECT Dept, COUNT(*) FROM Emp GROUP BY Dept
        HAVING COUNT(*) > 5;
```

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## Q24. What is a View?

Answer:

A view is a virtual table based on the result of a SQL query.

It does not store data physically.

Example:

```
CREATE VIEW EmpView AS
SELECT ID, Name, Salary FROM Employees;
```

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## Q25. What is the difference between DELETE, TRUNCATE and DROP?

Answer:

- DELETE → Removes specific rows. It is DML and can be rolled back.
- TRUNCATE → Removes all rows but keeps the table structure.  
It is DDL and faster than DELETE.
- DROP → Removes the entire table structure and data.  
It is DDL and cannot be rolled back.

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## Q26. ACID Properties in SQL?

Answer:

- Atomicity → All or nothing.
- Consistency → Data remains in a valid state.
- Isolation → Transactions do not affect each other.
- Durability → Once committed, data is permanent.

**Pro Tip:** Focus on writing queries on paper.

Practice more and more real-world queries.

--- End of SQL Interview Questions ---



## SQL Interview Questions - Page 6

Q27. What is an Index?

Answer:

An index is a database object that improves the speed of data retrieval operations on a table. It works like an index in a book.

Q28. What are the types of Indexes?

Answer:

- PRIMARY INDEX → Automatically created on Primary Key.
- UNIQUE INDEX → Automatically created on UNIQUE constraint.
- CLUSTERED INDEX → Sorts and stores the data rows.
- NON-CLUSTERED INDEX → Stores a separate structure with pointers.

Q29. What is the difference between CHAR and VARCHAR?

Answer:

- CHAR has fixed length and is space padded.
- VARCHAR has variable length and uses only required space.
- Use CHAR for fixed-length data, VARCHAR for variable-length data.

Example: CHAR(10) → 'A' stores as 'A\_\_\_\_\_ ' (10 chars)

VARCHAR(10) → 'A' stores as 'A' (1 char)

Q30. What is a Subquery vs a JOIN?

Answer:

- Subquery: A query inside another query.
- JOIN: Combines rows from two or more tables.
- Subqueries are easier for small queries,  
JOINS are more efficient for large datasets.

Q31. What is a Self Join?

Answer:

A self join is a regular join but the table is joined with itself.

It is useful for comparing rows within the same table.

Example: Find employees and their managers from the same table.

Q32. What is the difference between UNION and INTERSECT?

Answer:

- UNION → Returns all unique rows from both queries.
- INTERSECT → Returns only the common rows present in both queries.

Q33. What is the difference between EXISTS and IN?

Answer:

- EXISTS checks for the existence of any row returned by a subquery.
- IN checks if a value matches any value returned by a subquery.
- EXISTS is generally faster for correlated subqueries.

Q34. What is a Correlated Subquery?

Answer: A correlated subquery is a subquery that depends on the outer query because it references columns from the outer query.

Example: 

```
SELECT Name FROM Employees e1
WHERE Salary > (SELECT AVG(Salary) FROM Employees e2
WHERE e1.DeptID = e2.DeptID);
```

**Pro Tip:** Practice writing queries for real-world scenarios.  
The more you write, the better you get! ☆

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# SQL Interview Questions - Page 8

Q47. What is the difference between INNER JOIN and OUTER JOIN?

Answer:

- INNER JOIN → Returns only matching rows between tables.
- OUTER JOIN → Returns matching rows + non-matching rows from one or both tables.

Q48. What are the types of OUTER JOIN?

Answer:

- LEFT OUTER JOIN → All rows from left table + matching from right.
- RIGHT OUTER JOIN → All rows from right table + matching from left.
- FULL OUTER JOIN → All rows from both tables (match or no match).

Q49. What is NULLIF() function?

Answer:

NULLIF(expr1, expr2) returns NULL if expr1 = expr2, otherwise it returns expr1.

Example:

```
SELECT NULLIF(10, 10) → NULL
SELECT NULLIF(10, 20) → 10
```

Q50. What is COALESCE() function?

Answer:

COALESCE() returns the first non-NULL value in the list of expressions.

Example:

```
SELECT COALESCE(NULL, NULL, 'SQL', 'DB') → 'SQL'
```

Q51. What is the difference between TRUNC() and ROUND()?

Answer:

- TRUNC(n, d) → Truncates the number to d decimal places (no rounding).
- ROUND(n, d) → Rounds the number to d decimal places.

Example:

| Number   | TRUNC(n, 2) | ROUND(n, 2) |
|----------|-------------|-------------|
| 123.4567 | 123.45      | 123.46      |
| 123.444  | 123.44      | 123.44      |
| 123.445  | 123.44      | 123.45      |

Q52. What is a Stored Procedure?

Answer:

A stored procedure is a precompiled SQL code that can accept input parameters, perform operations and return a result.

Example:

```
CREATE PROCEDURE GetEmpByDept (IN deptId INT)
BEGIN
    SELECT * FROM Employees WHERE DeptID = deptId;
END;
```

Q53. What is a Trigger?

Answer:

A trigger is a special type of stored procedure that automatically executes in response to an event (INSERT, UPDATE, DELETE) on a table.

Example:

```
CREATE TRIGGER trg_emp_insert
AFTER INSERT ON Employees
FOR EACH ROW
BEGIN
    INSERT INTO EmpLog (EmpID, Action, ActionDate)
    VALUES (NEW.EmpID, 'Inserted', NOW());
END;
```

**Pro Tip:** Practice writing queries on real-world scenarios.  
The more you practice, the more confident you become! ★